

Feature Engineering - Study Notes

Why it matters

The quality of the features often matters more than the choice of algorithm. Feature engineering is the process of cleaning, transforming and creating input variables so that a model can learn from them more easily.

Missing values

If only a few rows or columns are affected, they can be dropped. Otherwise impute the missing values: use the mean or median for numeric features and the most frequent value for categorical features. The median is more robust to outliers than the mean. Adding a missing-value indicator column can also help.

Feature scaling

Standardization, also called z-score scaling, subtracts the mean and divides by the standard deviation so the feature has mean 0 and standard deviation 1. Min-max scaling rescales a feature to the range 0 to 1. Scaling is needed for distance-based and gradient-based methods such as k-NN, SVM, k-means, PCA and neural networks. Tree-based models do not need scaling.

Encoding categorical features

One-hot encoding creates one binary column for each category. For example, a color feature with the values red, green and blue becomes three columns. Ordinal encoding maps ordered categories such as small, medium and large to integers. For features with very many categories, target encoding or learned embeddings are often used instead.

Data leakage

Data leakage happens when information from the validation or test data leaks into training, which gives over-optimistic results. A typical example is fitting a scaler on the whole dataset before splitting it. To avoid leakage, split the data first, fit every transformer on the training set only, and then apply it to the validation and test sets. Scikit-learn Pipelines make this easy.

Class imbalance

When one class is rare, use class weights so that mistakes on the rare class cost more, oversample the minority class (for example with SMOTE), or undersample the majority class.

Feature selection

Feature selection removes irrelevant or redundant features. Common approaches are filtering highly correlated features, using L1 regularization, and using the importance scores of tree-based models.